

From Four Axioms to the Cosmological Constant: Causal Closure of the Chain C1–C4 to Λ

A Synthesis and Computational Verification

PDL Document D53

Cédric Laubscher

Independent Researcher, Switzerland

cedriclaubscher.ch

May 2026

Abstract

This document is a synthesis and computational verification of the causal chain leading from the four combinatorial axioms C1–C4 of the Projective Dynamic Logo (PDL) programme to the cosmological constant Λ . No new mathematical result is established here: all theorems are proved in the cited documents. The contribution of D53 is to present the complete chain in a single, self-contained text, and to provide an independent numerical verification of the key formula — the cosmological leakage constant C — at a precision of 0.41 ppm relative to the value published in D51 [1].

The chain traverses fifty-seven orders of magnitude in physical scale: from four axioms on finite signed graphs, without presupposing spacetime, particles, or fields, to the cosmological constant $\Lambda_{\text{PDL}} \approx 1.089 \times 10^{-52} \text{ m}^{-2}$. Each step is a theorem; no free parameter enters beyond the sole irreducible PDL–QCD interface parameter $\Delta m_{\text{iso}} = m_d - m_u$, which plays no role in the cosmological derivation. The smallness of Λ is not a fine-tuning: it is the arithmetic consequence of three independent leakage suppressions, each raised to a prime power imposed by the non-decomposability axiom C3.

Contents

1	Introduction	3
1.1	The scale of the result	3
2	The four axioms and their immediate consequences	4
2.1	Axioms C1–C4	4
2.2	The minimal closure is K_4	4
2.3	The proton quintuplet	4
3	The causal chain from C1–C4 to Λ	4
3.1	Overview	4
3.2	Step 1: $\beta_1(K_4) = 3$ forces three independent leakage cycles	5
3.3	Step 2: C3 forces the cycle lengths to be prime	5
3.4	Step 3: The quintuplet determines the three prime indices	5
3.5	Step 4: The three leakage base quantities	5
3.6	Step 5: The cosmological leakage constant C	6
3.7	Step 6: The cosmological constant	6
4	Numerical verification	6
4.1	Cell 1 — Fundamental PDL parameters	7
4.2	Cell 2 — Prime indices and exponents	7
4.3	Cell 3 — C , Λ_{PDL} , and robustness	8
4.4	Numerical results summary	9
4.5	Robustness of the prime structure	9
5	Epistemic status	10
5.1	What D53 does not claim	11
6	Open problems at the cosmological frontier	11
7	Conclusion	11

1 Introduction

The cosmological constant problem is one of the deepest open questions in theoretical physics. In the standard quantum field theory framework, the vacuum energy density is estimated at 10^{120} times the observed value of Λ . No satisfactory mechanism for this suppression has been established.

The PDL programme approaches the problem from the opposite direction. Rather than starting from a quantum field theory and asking why the vacuum energy is small, it starts from four combinatorial axioms on finite signed graphs and derives Λ as an unconditional consequence. The smallness of Λ is then not a mystery to be explained away, but a theorem to be proved.

Documents D51 [1] and D52 [2] together establish this derivation. D51 identifies the prime structure of the leakage exponents from the topology of K_4 and the pulsation axiom C1. D52 formally identifies the three leakage base quantities with the three orthogonal sectors of the proton relational architecture. The present document, D53, serves three purposes. First, it presents the complete causal chain from C1–C4 to Λ in a single linear narrative, with explicit reference to the source document for each step. Second, it provides a fully self-contained numerical verification in Python, reproduced verbatim, which an independent reader can execute to confirm the result. Third, it situates the result epistemically, identifying what is an unconditional theorem, what is a verified numerical identification, and what remains open.

1.1 The scale of the result

The chain spans the following conceptual distances:

- **Combinatorial layer:** Four axioms C1–C4 on finite signed graphs. No metric, no field, no spacetime.
- **Structural layer:** The minimal admissible closure is K_4 , the complete graph on four vertices. The proton is the minimal hierarchical composite, uniquely characterised by the quintuplet $(n_u, n_d, R_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10\,087, 11\,017)$.
- **Topological layer:** The first Betti number $\beta_1(K_4) = 3$ imposes exactly three independent leakage cycles.
- **Arithmetic layer:** Axiom C3 (non-decomposability) forces each leakage cycle length to be prime. The three primes are $p_{168} = 997$, $p_9 = 23$, $p_{19} = 67$, where the indices are determined by the quintuplet.
- **Physical layer:** The cosmological constant $\Lambda \approx 1.089 \times 10^{-52} \text{ m}^{-2}$ follows from the product of three suppression factors, each a rational function of the quintuplet raised to a prime power.

The ratio of the proton Compton wavelength squared (the natural PDL length scale) to Λ^{-1} (the cosmological length scale) is approximately 10^{57} . The derivation crosses this gap without invoking any intermediate energy scale, anthropic argument, or adjustable parameter.

2 The four axioms and their immediate consequences

2.1 Axioms C1–C4

The PDL programme rests on four axioms on finite signed graphs [3].

C1 Binary pulsation. Every closure undergoes a binary alternation between two complementary states. The minimal pulsation has exactly two branches.

C2 Relational completeness. Every pair of nodes is connected by at least one signed edge. No node is isolated.

C3 Non-decomposability. The closure cannot be partitioned into two independent sub-closures. Every subset is coupled to its complement.

C4 Minimality. The closure is the smallest structure satisfying C1–C3.

2.2 The minimal closure is K_4

Theorem 2.1 (Uniqueness of K_4 , D16a [4]). *The unique finite signed graph satisfying axioms C1–C4 is the complete graph K_4 on four vertices and six edges, with $R_e = 6$ relational degrees of freedom.*

This result is the foundation of the entire programme. K_4 is not postulated; it is the unique consequence of the four axioms.

2.3 The proton quintuplet

The proton is the minimal hierarchical composite closure admissible under C1–C4 [5, 6]. Its relational architecture is uniquely characterised by the integer quintuplet:

$$(n_u, n_d, R_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10\,087, 11\,017), \quad (1)$$

where n_u and n_d are the up- and down-valence core counts, R_{val} is the valence relational budget, R_{sea} is the sea budget, and $R_{\text{tot}} = R_{\text{val}} + R_{\text{sea}}$ is the total relational budget.

3 The causal chain from C1–C4 to Λ

3.1 Overview

The complete chain is:

$$\text{C1–C4} \xrightarrow{\text{D16a}} K_4 \xrightarrow{\text{D23}} \beta_1 = 3 \xrightarrow{\text{D51}} (p_{k_1}, p_{k_2}, p_{k_3}) \xrightarrow{\text{D52}} (\text{bases}) \xrightarrow{\text{D51}} C \xrightarrow{\text{D48}} \Lambda_{\text{PDL}}. \quad (2)$$

Each arrow is an established theorem. We now trace each step.

3.2 Step 1: $\beta_1(K_4) = 3$ forces three independent leakage cycles

Theorem 3.1 (Three leakage cycles, D23 [7]). *The first Betti number of the 1-skeleton of K_4 satisfies $\beta_1(K_4) = 3$. Consequently, there exist exactly three independent leakage cycles, corresponding to three orthogonal directions of relational incoherence.*

The Betti number $\beta_1 = |E| - |V| + 1 = 6 - 4 + 1 = 3$ is a combinatorial identity. The identification of these three cycles with leakage directions follows from the decomposition of the coherence stress-energy tensor in D48 [8].

3.3 Step 2: C3 forces the cycle lengths to be prime

Theorem 3.2 (Primality of leakage cycles, D51 [1]). *Under the conjunction of C1 (binary pulsation) and C3 (non-decomposability), a leakage cycle of length N is irreducible if and only if N is prime. Every composite length admits a factorisation that violates C3.*

This is the key structural constraint. The prime exponents in the formula for C are not fitted to the data: they are the unique lengths consistent with non-decomposability.

3.4 Step 3: The quintuplet determines the three prime indices

Theorem 3.3 (Index formula, D51 [1]). *The three leakage cycle lengths are counted by the following exact combinatorial identities in the quintuplet parameters:*

$$k_1 = n_d - n_u + R_e - 1 = 28 - 24 + 6 - 1 = 9, \quad (3)$$

$$k_2 = n_u - R_e + 1 = 24 - 6 + 1 = 19, \quad (4)$$

$$k_3 = R_e \cdot n_d = 6 \times 28 = 168. \quad (5)$$

The completeness identity $k_1 + k_2 = n_d = 28$ holds exactly. The three prime exponents are $p_{k_1} = p_9 = 23$, $p_{k_2} = p_{19} = 67$, and $p_{k_3} = p_{168} = 997$, where p_k denotes the k -th prime number.

3.5 Step 4: The three leakage base quantities

The three orthogonal leakage directions correspond to three natural base quantities, each an unconditional theorem of the PDL programme [2]:

1. **Surface leakage base** $(1 - \kappa)$, where $\kappa = R_{\text{surf}}/R_{\text{tot}} = 310\varphi/11\,017 \in \mathbb{Q}(\sqrt{5})$ is the surface-to-total coherence ratio (D42 [9]). This base quantifies the fraction of relational activity that fails to contribute to the active surface per pulsation cycle.
2. **Valence leakage base** $R_{\text{val}}/R_{\text{tot}} = 930/11\,017$, the fraction of the total relational budget that is directly available for valence coupling (D29 [10]).
3. **Coupling leakage base** $(1 - \eta_L)$, where $\eta_L = \varepsilon_G^B \in \mathbb{Q}(\sqrt{5})$ is the gravitational coupling leakage parameter established in D30 [11]:

$$\eta_L = \frac{\kappa \cdot C_0}{6 + \kappa} \cdot \left(1 + \frac{2}{R_{\text{tot}}}\right), \quad C_0 = \frac{n_u^2 - 1}{n_u^2} = \frac{575}{576}. \quad (6)$$

This base quantifies the per-level attenuation of the gravitational coupling across the 18-step hierarchical filter.

3.6 Step 5: The cosmological leakage constant C

Theorem 3.4 (Cosmological leakage constant, D51 [1] + D52 [2]). *The constant C in the leakage term of the PDL coherence stress-energy tensor (D48 [8]) is:*

$$C = (1 - \kappa)^{997} \times \left(\frac{R_{\text{val}}}{R_{\text{tot}}} \right)^{23} \times (1 - \eta_L)^{67}. \quad (7)$$

No free parameter enters (7). All three bases and all three exponents are unconditional theorems of C1–C4 via the quintuplet.

3.7 Step 6: The cosmological constant

The cosmological leakage term in the PDL Einstein field equation (D48 [8], D35 [12]) is:

$$C_{\mu\nu}^{(\text{leak})} = -C \cdot \eta_L^{18} \cdot \left(\frac{m_p c}{\hbar} \right)^2 g_{\mu\nu}. \quad (8)$$

Identifying this term with the cosmological constant contribution $-\Lambda g_{\mu\nu}$ gives:

$$\Lambda_{\text{PDL}} = C \cdot \eta_L^{18} \cdot \left(\frac{m_p c}{\hbar} \right)^2. \quad (9)$$

Evaluating (9) with the algebraic value $\eta_L = 0.00751927 \dots \in \mathbb{Q}(\sqrt{5})$ and CODATA 2022 values for m_p , c , $\hbar$ yields:

$$\Lambda_{\text{PDL}} \approx 1.089\,009 \times 10^{-52} \text{ m}^{-2}, \quad (10)$$

in agreement with the Planck 2020 observational value $\Lambda_{\text{obs}} = 1.089 \times 10^{-52} \text{ m}^{-2}$ [13] within the measurement uncertainty of $\pm 1.7\%$.

Remark 3.5 (Why Λ is small). The smallness of $\Lambda \approx 10^{-52} \text{ m}^{-2}$ is not a fine-tuning mystery within the PDL framework. It is the product of three quantities each close to 1, raised to large prime powers: $0.9545^{997} \approx 10^{-20}$, $(930/11\,017)^{23} \approx 10^{-26}$, and $0.9925^{67} \approx 10^{-0.4}$. The total is $C \approx 10^{-46}$, and $\eta_L^{18} \approx 10^{-38}$ provides the remaining suppression relative to the Planck scale. Each large exponent is a prime forced by C3. The universe's residual incoherence accumulates over $997 + 23 + 67 = 1087$ irreducible leakage events across three independent sectors.

4 Numerical verification

The following Python code, executable in Google Colab with standard scientific libraries, provides an independent numerical verification of the chain described in Section 3. Three self-contained cells are presented in logical order; each cell repeats all necessary imports.

4.1 Cell 1 — Fundamental PDL parameters

Cell 1: Fundamental PDL parameters (50-decimal precision)

```
from sympy import sqrt, Integer, N as Neval
from mpmath import mp, mpf, nstr, power

mp.dps = 50

# Quintuplet (unconditional theorems)
n_u, n_d, R_e      = 24, 28, 6
R_val, R_sea, R_tot = 930, 10087, 11017

phi  = mpf(str(Neval((1 + sqrt(5)) / 2, 60)))
kappa = mpf(310) * phi / mpf(11017)

# eta_L = varepsilon_G^B (algebraic, D30, in Q(sqrt(5)))
C_nulo = (mpf(n_u)**2 - 1) / mpf(n_u)**2 # 575/576
eta_L   = kappa * C_nulo / (6 + kappa) * (1 + mpf(2)/mpf(R_tot))

r_val  = mpf(R_val) / mpf(R_tot)

assert abs(kappa - mpf(310)*phi/mpf(11017)) < mpf('1e-48')
assert R_val + R_sea == R_tot
assert 0 < float(eta_L) < 1

# Results: kappa = 0.045528777..., eta_L = 0.007519273...
```

4.2 Cell 2 — Prime indices and exponents

Cell 2: Prime indices and exponents

```
from sympy import prime, isprime

n_u, n_d, R_e = 24, 28, 6

k1 = n_d - n_u + R_e - 1 # = 9
k2 = n_u - R_e + 1       # = 19
k3 = R_e * n_d           # = 168

assert k1 + k2 == n_d     # completeness identity

p_k1 = int(prime(k1))     # p_9  = 23
p_k2 = int(prime(k2))     # p_19 = 67
p_k3 = int(prime(k3))     # p_168 = 997

for k, p in [(k1, p_k1), (k2, p_k2), (k3, p_k3)]:
    assert isprime(p)

# All three exponents confirmed prime: 23, 67, 997
```

4.3 Cell 3 — C , Λ_{PDL} , and robustness

Cell 3: C , Λ_{PDL} , deviation, and table of exclusion

```
from sympy import sqrt, N as Neval, prime
from mpmath import mp, mpf, log, exp, fabs, nstr, power

mp.dps = 50

# Quintuplet and parameters (repeated for self-containment)
n_u, n_d, R_e      = 24, 28, 6
R_val, R_sea, R_tot = 930, 10087, 11017
phi      = mpf(str(Neval((1 + sqrt(5)) / 2, 60)))
kappa    = mpf(310) * phi / mpf(11017)
C_nulo   = (mpf(n_u)**2 - 1) / mpf(n_u)**2
eta_L    = kappa * C_nulo / (6 + kappa) * (1 + mpf(2)/mpf(R_tot))
r_val    = mpf(R_val) / mpf(R_tot)

p_k1, p_k2, p_k3 = 23, 67, 997

b_surf = 1 - kappa
b_val  = r_val
b_coup = 1 - eta_L

log_C = p_k3*log(b_surf) + p_k1*log(b_val) + p_k2*log(b_coup)
C_PDL = exp(log_C)

# CODATA 2022
m_p  = mpf('1.67262192369e-27') # kg
c    = mpf('299792458')         # m/s
hbar = mpf('1.054571817e-34')   # J s
mp_c_hbar_sq = (m_p * c / hbar)**2
eta_L_18     = power(eta_L, 18)

Lambda_PDL = C_PDL * eta_L_18 * mp_c_hbar_sq

# Reference: C_target published in D51
C_target_D51 = mpf('8.1579491e-46')
dev_ppm = fabs(C_PDL - C_target_D51) / C_target_D51 * mpf('1e6')

# Planck 2020
Lambda_obs = mpf('1.089e-52')
C_target_Planck = Lambda_obs / (eta_L_18 * mp_c_hbar_sq)

# Results:
# C_PDL      = 8.15794572e-46
# C_target (D51) = 8.15794910e-46 -> deviation 0.41 ppm
# Lambda_PDL   = 1.08901e-52 m^-2 -> ratio 1.0000082
```


4.4 Numerical results summary

Table 1 summarises the key numerical results of the three cells.

Table 1: Numerical verification results (50-decimal precision, mpmath). The structural deviation of 0.41 ppm is the deviation of C_{PDL} from the value C_{target} published in D51 [1]. The apparent deviation of 8.17 ppm relative to the Planck 2020 value arises entirely from an 8.6 ppm metrological offset in $(m_p c/\hbar)^2$ between the CODATA values used in D51 and CODATA 2022 — it carries no structural significance.

Quantity	Expression	Value
κ	$310\varphi/11\,017$	$0.045\,528\,777\dots$
η_L	$\kappa C_0/(6 + \kappa) \cdot (1 + 2/R_{\text{tot}})$	$0.007\,519\,273\dots$
$R_{\text{val}}/R_{\text{tot}}$	$930/11\,017$	$0.084\,415\dots$
k_1, k_2, k_3	9, 19, 168	exact integers
$p_{k_1}, p_{k_2}, p_{k_3}$	p_9, p_{19}, p_{168}	23, 67, 997
C_{PDL}	$(1 - \kappa)^{997} (R_{\text{val}}/R_{\text{tot}})^{23} (1 - \eta_L)^{67}$	$8.157\,946 \times 10^{-46}$
C_{target} (D51)	published value	$8.157\,949 \times 10^{-46}$
Structural deviation	$ C_{\text{PDL}} - C_{\text{target}} /C_{\text{target}}$	0.41 ppm
η_L^{18}	$= \alpha_G$ (Gate 2)	5.904×10^{-39}
$(m_p c/\hbar)^2$	CODATA 2022	$2.261 \times 10^{31} \text{ m}^{-2}$
Λ_{PDL}	$C \cdot \eta_L^{18} \cdot (m_p c/\hbar)^2$	$1.089\,009 \times 10^{-52} \text{ m}^{-2}$
Λ_{obs}	Planck 2020 [13]	$1.089 \times 10^{-52} \text{ m}^{-2}$
$\Lambda_{\text{PDL}}/\Lambda_{\text{obs}}$		1.000 008

4.5 Robustness of the prime structure

The primality of the exponents is structurally necessary, not incidental. Table 2 reproduces the robustness test of D51 [1]: replacing any prime exponent by its nearest composite neighbour increases the deviation from C_{target} by at least four orders of magnitude.

Table 2: Robustness of the prime exponents. Replacing any prime by its nearest composite neighbour degrades the agreement with C_{target} (D51) by at least four orders of magnitude. The prime structure is load-bearing.

Substitution	Exponents	Deviation from C_{target}
Reference (all primes)	(997, 23, 67)	0.41 ppm
$997 \rightarrow 996 = 4 \times 249$	(996, 23, 67)	47 700 ppm
$997 \rightarrow 998 = 2 \times 499$	(998, 23, 67)	45 529 ppm
$23 \rightarrow 22 = 2 \times 11$	(997, 22, 67)	10 846 235 ppm
$23 \rightarrow 24 = 2^3 \times 3$	(997, 24, 67)	915 585 ppm
$67 \rightarrow 66 = 2 \times 3 \times 11$	(997, 23, 66)	7 576 ppm
$67 \rightarrow 68 = 4 \times 17$	(997, 23, 68)	7 519 ppm

5 Epistemic status

Table 3: Epistemic status of the principal results of D53. All results originate in the cited documents; D53 synthesises and verifies them independently.

Result	Status	Source
$C1-C4 \Rightarrow K_4$ unique	Theorem	D16a
$\beta_1(K_4) = 3$	Theorem	D23
Three orthogonal leakage sectors	Theorem	D29, D42, D17
C3 forces prime cycle lengths	Theorem	D51
Index formula k_1, k_2, k_3 from quintuplet	Theorem	D51
Completeness identity $k_1 + k_2 = n_d$	Theorem	D51
$(1 - \kappa) = \text{surface leakage base}$	Theorem	D42
$R_{\text{val}}/R_{\text{tot}} = \text{valence leakage base}$	Theorem	D29
$(1 - \eta_L) = \text{coupling leakage base}$	Theorem	D17, D30
Algebraic independence of three bases	Verified	D52
Uniqueness by exhaustive exclusion	Verified	D52
Formula (7) as unconditional theorem	Theorem	D51 + D52
C_{PDL} vs C_{target} (D51): 0.41 ppm	Verified	D53 (this document)
$\Lambda_{\text{PDL}} \approx 1.089 \times 10^{-52} \text{ m}^{-2}$	Prediction	D51, D52, D53
Causal chain $C1-C4 \rightarrow \Lambda$ complete	Theorem	D16a–D52

5.1 What D53 does not claim

Epistemic precision requires an explicit statement of what is not established here. Three points deserve mention.

First, the identification of the three leakage bases with the three sectors is proved in D52 by a combination of unconditional structural theorems and exhaustive exclusion of alternatives [2]. This constitutes a rigorous proof by elimination. A direct algebraic construction of the bijection from C1–C4 alone, without the exclusion step, would strengthen the result further, but is not required for the theorem to hold.

Second, the comparison with Λ_{obs} involves CODATA values for m_p , c , and $\hbar$. The formula (9) is structurally exact; the numerical comparison inherits the metrological uncertainty of these constants (approximately ± 12 ppm for m_p and ± 26 ppm for G). The 0.41 ppm deviation is below this noise floor and is therefore consistent with exact structural agreement.

Third, the sole irreducible external parameter of the PDL programme, $\Delta m_{\text{iso}} = m_d - m_u$, plays no role in the cosmological derivation. The chain C1–C4 $\rightarrow \Lambda$ is parameter-free.

6 Open problems at the cosmological frontier

The closure of the chain C1–C4 $\rightarrow \Lambda$ does not exhaust the cosmological questions accessible to PDL. The following three problems remain open.

OP-pressure. The equation of state $w = P/(\rho_{\text{coh}} c^2)$ for the coherence fluid has not been derived from C1–C4. The pressure term in the coherence stress-energy tensor (D48 [8]) is constrained but not fully determined.

OP-London. The connection between the orbital component $C_{\mu\nu}^{(\text{orb})}$ and the London equation (D49 [14], D46 [15]) has been conjectured but not formally established as a theorem.

OP-CMB. The derivation of the CMB angular power spectrum C_ℓ from the PDL cosmological framework, using the $\sigma(N)$ -modified gravitational coupling, has not been attempted. The scalar spectral index $n_s \approx 0.965$ and the tensor-to-scalar ratio r are the primary targets.

7 Conclusion

The causal chain from four combinatorial axioms on finite signed graphs to the cosmological constant $\Lambda \approx 1.089 \times 10^{-52} \text{ m}^{-2}$ is complete. Each step is an established theorem of the PDL programme. No free parameter enters the derivation of Λ . The smallness of the cosmological constant is a combinatorial necessity: it is the product of three suppression factors, each raised to a prime power forced by the non-decomposability axiom C3.

The independent numerical verification presented in Section 4 confirms the formula to 0.41 ppm relative to the published value of D51 [1]. The remaining discrepancy relative to the Planck 2020 observational value is of metrological origin and carries no structural significance.

The cosmological constant Λ_{PDL} is an unconditional consequence of the four axioms C1–C4, via the chain (2). The derivation uses no free parameter, no spacetime manifold, no quantum field, and no anthropic reasoning. The chain is complete.

References

- [1] Cédric Laubscher. The cosmological leakage constant from the PDL axioms: A structural derivation of c and resolution of OP1-D35. Zenodo, 2026.
- [2] Cédric Laubscher. Formal identification of the three leakage bases as natural representatives of the three cycles of $\beta_1(k_4) = 3$: Completion of the proof of conjecture PDL-C. Zenodo, 2026.
- [3] Cédric Laubscher. The emergence of physical reality in projective dynamic logo (PDL). Zenodo, 2026.
- [4] Cédric Laubscher. Minimal stationary closures and the necessity of the $(4, 6)$ block in the PDL framework. Zenodo, 2026.
- [5] Cédric Laubscher. Combinatorial selection and local uniqueness of the proton architecture in the PDL framework. Zenodo, 2026.
- [6] Cédric Laubscher. Derivation of the sub-shell filling rates and the periodic table from the PDL axioms: Resolution of OP13 and OP14. Zenodo, 2026.
- [7] Cédric Laubscher. Topological constraints and the 18-step hierarchy in the PDL framework. Zenodo, 2026.
- [8] Cédric Laubscher. Derivation of the coherence stress-energy tensor from the PDL axioms: Explicit form of $\mathcal{C}_{\text{coh}}$ and resolution of OP2 (D35). Zenodo, 2026.
- [9] Cédric Laubscher. Derivation of H3 from axioms C1–C4: the equiparticipation lemma and the closure of causality. Zenodo, 2026.
- [10] Cédric Laubscher. Gate 1: Derivation of 155/11017 from the PDL axioms. Zenodo, 2026.
- [11] Cédric Laubscher. Gate 2: Proof of the ε_g^b conjecture and full derivation of Newton's constant. Zenodo, 2026.
- [12] Cédric Laubscher. Quantitative Einstein equation from $g_{\text{eff}}(n) = \sigma(n) \cdot g_{\text{pdl}}$ in the PDL framework. Zenodo, 2026.
- [13] Planck Collaboration. Planck 2018 results. VI. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020.
- [14] Cédric Laubscher. Derivation of the London equation from the PDL axioms: Resolution of OP-London. Zenodo, 2026.
- [15] Cédric Laubscher. Born's rule level 2: Derivation of U(1) phase freedom from the k_4 pulsation structure in the PDL framework. Zenodo, 2026.